

- 2** Use a double-angle formula to write  $\cos^2 x$  in terms of  $\cos 2x$ , and state the corresponding formula for  $\sin^2 x$ .

- 6** Solve  $\tan^2 x = 3$  on the interval  $[0, 2\pi)$ .

- 8** Given  $\cos \theta = \frac{7}{25}$  with  $0 < \theta < \frac{\pi}{2}$ , find  $\sin \frac{\theta}{2}$  exactly.

- 10** Solve  $\sin 2x - \cos x = 0$  on the interval  $[0, 2\pi)$ .

- 11** A projectile's range is given by  $R(\theta) = \frac{v^2 \sin 2\theta}{g}$ . Using the double-angle formula, explain why the maximum range occurs at  $\theta = 45^\circ$ .

- 12 Given  $\sin \theta = \frac{1}{3}$  with  $\theta$  in Quadrant I, find  $\cos 2\theta$  and  $\sin 2\theta$  exactly.
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### More equation practice

- 13 Solve  $4\sin^2 x - 3 = 0$  on the interval  $[0, 2\pi)$ .
- 14 Solve  $\cos 2x = \cos x$  on the interval  $[0, 2\pi)$ , using  $\cos 2x = 2\cos^2 x - 1$ .